

# Paper Selection Proposal

## Single-Source Shortest Paths with Negative Real Weights in $O(mn^{8/9})$ Time

Team Members: Hussain Umer Farooqui, Hussain Mustansir

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### Paper Details

- **Title:** Single-Source Shortest Paths with Negative Real Weights in  $O(mn^{8/9})$  Time
- **Authors:** Jeremy T. Fineman
- **Conference:** Submitted to STOC (2024)
- **Year:** 2024
- **DOI/Link:** <https://dl.acm.org/doi/10.1145/3618260.3649614>

### Summary

This paper presents a randomized algorithm for solving the Single-Source Shortest Paths (SSSP) problem on directed graphs with real edge weights, both positive and negative. The key contribution is an improved running time of  $O(mn^{8/9})$ , where  $m$  is the number of edges and  $n$  the number of vertices, which improves upon the Bellman-Ford algorithm's  $O(mn)$  time by using Dijkstra Algorithm.

### Justification

This paper's novel time complexity improvement for SSSP is relevant to graph-based optimization problems, including network analysis and routing. It offers a practical improvement for graphs with real weights and negative edges. And it aligns with one of the topics in our course syllabus.

### Implementation Feasibility

The paper provides detailed theoretical descriptions of the algorithm. While no direct code is available, existing algorithms like Bellman-Ford and Dijkstra can serve as a foundation for implementation.

### Team Responsibilities

Both of us will contribute equally to every task.